

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET D — ASSERTION & REASONING SPECIAL****CHAPTER 02: SHAPING OF THE EARTH'S SURFACE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-D-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET D - Assertion & Reasoning Special):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Concepts & Theoretical Principles.
5. **Section B (Q11 – Q18):** Evidentiary Proofs, Case Inferences & Historical/Geographical Mechanisms.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Institutional/Analytical Complexities.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10) [10 MARKS]**Q.1**

Assertion (A): Endogenic geomorphic forces derive their energy primarily from radioactivity, rotational friction, and primordial heat within the Earth.

Reason (R): Endogenic forces induce diastrophism and volcanism, leading to the elevation and deformation of the Earth's lithosphere.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): The Himalayas are classified as young fold mountains formed by the collision of tectonic plates.

Reason (R): The Indian plate collided with the Eurasian plate at a convergent boundary, crumpling marine sediments of the Tethys geosyncline into massive folds.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): Seismic S-waves can propagate freely through the Earth's liquid outer core.

Reason (R): Transverse shear waves (S-waves) require shear elasticity and cannot propagate through fluids lacking shear rigidity.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): The epicentre of an earthquake experiences the earliest arrival of seismic body waves.

Reason (R): The epicentre is the point on the Earth's surface situated perpendicularly above the subterranean hypocentre (focus).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): Mid-oceanic ridges are formed along divergent plate boundaries where tectonic plates pull apart.

Reason (R): Upwelling basaltic magma from the asthenosphere cools and solidifies along the rift zone, continuously generating new oceanic crust.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): Transform faults neither create nor destroy lithospheric crust.

Reason (R): At transform plate boundaries, lithospheric plates slide horizontally past one another along strike-slip fractures.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.7

Assertion (A): Chemical weathering proceeds at the most rapid rate in hyper-arid hot desert environments.

Reason (R): Chemical weathering processes such as carbonation, hydration, and oxidation require liquid water as a primary solvent and reaction medium.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.8

Assertion (A): Frost wedging is a potent mechanical weathering agent in high-altitude periglacial mountain regions.

Reason (R): Water expands by approximately 9% in volume when freezing into ice, exerting immense tensile pressure within rock joints.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.9

Assertion (A): Weathering and erosion are identical geomorphic terms that can be used interchangeably.

Reason (R): Weathering involves the in-situ disintegration of rocks without transport, whereas erosion involves the detachment and dynamic transportation of debris.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.10

Assertion (A): Deep-sea oceanic trenches are located predominantly along convergent subduction plate boundaries.

Reason (R): When a dense oceanic plate collides with a lighter continental or island arc plate, the oceanic slab flexes and subducts into the asthenosphere.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EVIDENTIARY PROOFS & MECHANISMS (Q.11 TO Q.18) [8 MARKS]**Q.11**

Assertion (A): A river forms a V-shaped valley during its youthful stage in mountainous terrain.

Reason (R): In its upper course, a steep hydraulic gradient causes vertical downward erosion to vastly exceed lateral bank erosion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): Oxbow lakes are formed along the upper youthful course of mountain torrents.

Reason (R): Oxbow lakes originate on low-gradient mature floodplains when meander loops are cut off during seasonal flood discharges.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): Glaciers carve distinct U-shaped valleys with flat valley floors and steep vertical walls.

Reason (R): Glacial ice fills preexisting river valleys and erodes simultaneously across the base and lateral rock margins via plucking and abrasion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): Cirques are amphitheatre-like hollows carved by glaciers high on mountain slopes.

Reason (R): Freeze-thaw nivation and rotational basal ice plucking excavate deep bedrock depressions below mountain crests.

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| <p>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).</p> <p>(C) Assertion (A) is true but Reason (R) is false.</p> | <p>(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).</p> <p>(D) Assertion (A) is false but Reason (R) is true.</p> |
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Q.15

Assertion (A): Mushroom rocks (pedestal rocks) in arid deserts exhibit narrow stems and wide upper caps.

Reason (R): Wind-driven sand abrasion is concentrated primarily within the first 1 to 2 metres above the ground surface.

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| <p>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).</p> <p>(C) Assertion (A) is true but Reason (R) is false.</p> | <p>(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).</p> <p>(D) Assertion (A) is false but Reason (R) is true.</p> |
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Q.16

Assertion (A): Barchan sand dunes have their crescent tips or horns pointing against the prevailing wind direction.

Reason (R): Wind pushes sand up the gentle windward slope, allowing sand grains to avalanche down the steep leeward slip face, advancing the horns downwind.

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| <p>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).</p> <p>(C) Assertion (A) is true but Reason (R) is false.</p> | <p>(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).</p> <p>(D) Assertion (A) is false but Reason (R) is true.</p> |
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Q.17

Assertion (A): Alluvial fans develop at the base of mountain fronts where streams debouch onto flat plains.

Reason (R): An abrupt reduction in slope gradient causes a sudden drop in stream velocity and immediate deposition of coarse sediment loads.

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| <p>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).</p> <p>(C) Assertion (A) is true but Reason (R) is false.</p> | <p>(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).</p> <p>(D) Assertion (A) is false but Reason (R) is true.</p> |
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Q.18

Assertion (A): Deltas are formed at the river mouth when sediment deposition exceeds removal by ocean tides and currents.

Reason (R): River velocity approaches zero upon entering a standing body of water, triggering flocculation of clay and deposition of suspended bedload.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

SECTION C: ADVANCED NUANCES & INSTITUTIONAL COMPLEXITIES (Q.19 TO Q.25) [7 MARKS]**Q.19**

Assertion (A): Earthquake shadow zones exist for both P-waves and S-waves.

Reason (R): P-waves are refracted at the core-mantle boundary creating a shadow between 105° and 145°, while S-waves are blocked entirely by the liquid outer core beyond 105°.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.20

Assertion (A): Exfoliation domes form in granitic rock masses due to unloading and pressure release.

Reason (R): When overlying rock burdens are removed by long-term erosion, underlying plutonic rocks expand outward and spall along curved joints.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.21

Assertion (A): Biological weathering is caused exclusively by the physical growth of tree roots inside rock fissures.

Reason (R): Lichens, mosses, and burrowing organisms secrete organic chelating acids that chemically decompose rock silicate minerals.

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|---|---|
| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.22

Assertion (A): River meanders continually migrate laterally across their floodplains.

Reason (R): Centrifugal water acceleration causes higher velocity and erosion on the concave outer bank, while lower velocity causes deposition on the convex inner bank.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): Soil liquefaction during an earthquake occurs in well-consolidated bedrock foundations.

Reason (R): Liquefaction occurs when saturated loose sandy sediments lose shear strength under cyclic seismic shaking and behave like a fluid.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): A wave-cut platform is a depositional coastal landform composed of loose sand dunes.

Reason (R): Wave-cut platforms are flat erosional bedrock benches carved at the base of retreating sea cliffs by hydraulic wave action and abrasion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): The Pacific Rim is referred to as the 'Ring of Fire' due to intense volcanic and seismic activity.

Reason (R): The perimeter of the Pacific Ocean is bounded almost entirely by convergent plate boundaries characterized by subduction zones and deep ocean trenches.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & HISTORICAL/GEOGRAPHICAL CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Causal Rationale & Conceptual Explanation
Q.1	(A)	Assertion is true. Reason is true and explains how internal thermal energy drives tectonic landform creation.
Q.2	(A)	Assertion is true. Reason is true and explains the plate tectonic mechanism behind the Himalayan orogeny.
Q.3	(D)	Assertion is false: S-waves cannot travel through liquids. Reason is true: Transverse waves require shear strength, which fluids lack.
Q.4	(A)	Assertion is true. Reason is true and correctly explains why the epicentre is closest to the subterranean focus and feels the shock first.
Q.5	(A)	Assertion is true. Reason is true and provides the geophysical mechanism of seafloor spreading at divergent margins.
Q.6	(A)	Assertion is true. Reason is true and explains why strike-slip lateral movement is conservative with respect to crustal area.
Q.7	(D)	Assertion is false: Chemical weathering is slowest in dry deserts and fastest in warm, humid tropics. Reason is true: Water is essential for chemical weathering reactions.
Q.8	(A)	Assertion is true. Reason is true and explains the volumetric physics causing frost shattering in jointed rocks.
Q.9	(D)	Assertion is false: Weathering and erosion are distinct geomorphic processes. Reason is true: Weathering is static/in-situ while erosion involves transport by agents.
Q.10	(A)	Assertion is true. Reason is true and correctly explains the tectonic formation of trenches such as the Mariana Trench.
Q.11	(A)	Assertion is true. Reason is true and explains why dominantly vertical corrasion carves deep, narrow V-shaped gorges.
Q.12	(D)	Assertion is false: Oxbow lakes form on mature floodplains, not upper mountain reaches. Reason is true: Meander neck cutoffs happen on flat plains.
Q.13	(A)	Assertion is true. Reason is true and explains how glacial cross-sectional widening converts V-shaped gorges into U-shaped troughs.
Q.14	(A)	Assertion is true. Reason is true and describes the geomorphic excavating process of glacial cirque formation.
Q.15	(A)	Assertion is true. Reason is true and explains why differential aeolian saltation abrasion undercuts rock bases.
Q.16	(D)	Assertion is false: Barchan horns point in the direction of wind flow (downwind). Reason is true: Sand saltation advances the dune horns downwind.
Q.17	(A)	Assertion is true. Reason is true and explains the hydraulic deposition mechanics of alluvial fan formation.
Q.18	(A)	Assertion is true. Reason is true and explains the depositional equilibrium required for deltaic progradation.
Q.19	(A)	Assertion is true. Reason is true and provides the geophysical explanation for seismic wave shadow zone boundaries.
Q.20	(A)	Assertion is true. Reason is true and explains the mechanical pressure-release mechanism of sheeting and exfoliation.
Q.21	(D)	Assertion is false: Biological weathering involves both mechanical root wedging and biochemical acid secretion. Reason is true: Lichens produce chemical weathering acids.
Q.22	(A)	Assertion is true. Reason is true and correctly explains the hydrodynamic cause of lateral meander migration.

Q. No.	Correct	Causal Rationale & Conceptual Explanation
Q.23	(D)	Assertion is false: Liquefaction occurs in water-saturated loose sands/silts, not consolidated solid bedrock. Reason is true: Seismic pore-water pressure destroys sediment shear strength.
Q.24	(D)	Assertion is false: Wave-cut platforms are erosional bedrock benches, not sandy depositional features. Reason is true: Marine cliff retreat leaves behind planar wave-cut platforms.
Q.25	(A)	Assertion is true. Reason is true and explains why subduction volcanism and megathrust earthquakes circle the Pacific basin.